A fast and accurate class of carrier recovery schemes

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***Abstract*—This paper introduces a class of carrier recovery scheme based on simple trigonometric identities. We revisit carrier recovery schemes that have been in use for many years, and compare their relative performances using quantitative metrics. We show that trigonometric identities can be used to recover carriers faster and better than conventional control loop based schemes in special cases where there is a unique frequency locked relationship between the carriers and the pilot of the scheme. PLL and Costas loop based schemes have been carried over from analog systems and are now implemented on digital hardware without fully exploiting the advantages of digital systems, but we propose a simple scheme that can outperform these legacy methods in specific protocols, and be incorporated into modern receivers of any protocol having frequency-locked carriers of a pilot. We present results for the specific case of the FM-RDS protocol as an example.**

***Index Terms*—Synchronous detection, Signal processing, Phase locked loops, Amplitude modulation**

I. Introduction

THE synchronization of carriers at the receiver end of a communications system is a significant problem as it affects data recovery. In many communications systems, data is first modulated at the baseband by the double-sided suppressed-carrier (DSB-SC) method. In a general DSB-SC based method [1], a sinusoidal carrier, defined as

$$c(t) = cos(2\pi f_c t) \tag{1}$$

is multiplied with the lowpass message signal, $m(t)$, to produce a frequency-shifted version of the message signal that is centred at f_c. In the demodulation process, $c(t)$ is recovered and multiplied with the received signal, to shift the received signal back to baseband. The recovery of $c(t)$ thus plays a vital role in the demodulation. In general, phase-locked loops (PLL) or Costas loops are used for this purpose for a wide variety of protocols to track frequency or phase deviations that may occur in the carriers [2]–[5]. These methods are carried over from analog implementations onto modern digital systems. They are fundamentally limited by their stabilization time and error tracking capabilities. However, they are general-purpose schemes that work for a wide variety of protocols. This paper presents an accurate and immediate carrier recovery scheme for protocols that have frequency locked carriers of a pilot tone. We show that in such protocols, the scheme we present has distinct advantages over conventional carrier recovery schemes.

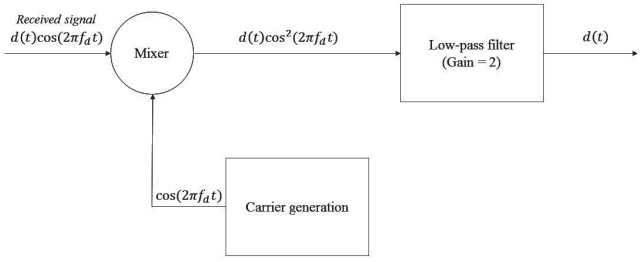


Fig. 1. Block diagram of ideal carrier recovery

Research on carrier recovery schemes is scarce in recent years, and to our knowledge this is the first treatment of the use of trigonometric identities for protocols that have carriers that are harmonics of a pilot. We present a scheme that does not require parameter tuning as in the case of conventional carrier recovery schemes such as the phase-locked loop (PLL) or Costas loops. The method is agnostic to the protocol as long as it has carriers that are harmonics of a recoverable pilot tone.

The paper is organized as follows: Carrier recovery is reviewed, the proposed scheme is presented, results and comparisons are presented by taking the case of the FM-RDS protocol as an example, and the results are discussed and conclusions drawn in the final section.

II. Carrier recovery

For a given message signal $d(t)$, where t represents time, the message signal is multiplied with a carrier signal $cos(2\pi f_d t)$ to generate

$$g(t) = d(t)cos(2\pi f_d t) \tag{2}$$

where f_d is a specified frequency. The spectrum of $g(t)$ is thus shifted to be centered at a frequency of f_d. Different signals can be sent at different frequencies by this method.

At the receiver end, the carriers are to be generated and multiplied with the signals again. The results are passed through low-pass filters to recover the message. The frequency and phase offsets in the carrier which occur during the transmission will cause differences between the generated carrier at the receiver end and the actual carrier used at the transmission end. The block diagram for a receiver system with ideal carrier synchronization is shown in Fig. 1.

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Fig. 2. Schematic of recovery of Nth harmonic carrier from pilot using a control loop

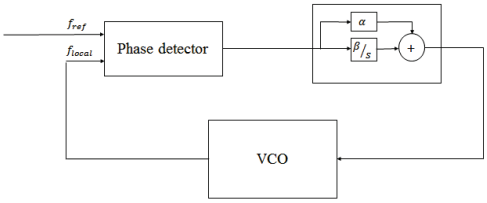


Fig. 3. Block diagram of a Phase locked loop

Many protocols have a pilot tone in addition to the message and the carriers. In the specific case where the carriers are harmonics of the pilot, the pilot is raised to the corresponding power, bandpassed, and then passed through a PLL or Costas loop. The scheme is shown in Fig. 2 for the carrier recovery in the general case when the carrier is the Nth harmonic of the pilot.

If there is a frequency offset of Δf and a phase offset of $\Delta\phi$ in the carrier used at the receiver end, the product of the mixer in the scheme shown in Figure 2 can be written as

$$\begin{aligned} r(t) &= d(t)cos(2\pi f_d t)cos(2\pi(f_d + \Delta f)t + \Delta\phi) \\ &= d(t)\frac{cos(2\pi\Delta f t + \Delta\phi) + cos(2\pi(2f_d + \Delta f)t + \Delta\phi)}{2} \end{aligned} \tag{3}$$

The second term will be removed by the low-pass filter. The first term will, however, remain and cause serious distortions and frequency shifts in the signal $d(t)$.

The design of the control loop is vital to the recovery in conventional schemes. Bellisio [6] recommends a PLL with loop filter

$$H(s) = \alpha + \frac{\beta}{s} \tag{4}$$

to be able to track both frequency and phase offsets. The parameters α and β have to be chosen by the designer, and the performance of the recovery scheme varies widely with different values of and within the stability range, as well as different types of PLLs. Besides this, the bandpass filter that is required before the PLL introduces a phase offset in the recovered carrier for any frequency offset in the pilot signal. The block diagram for a conventional second-order PLL following Bellisio's second-order filter scheme is shown in Figure 3.

A Costas loop [7] is similar to a PLL, and is also used for carrier recovery. For completeness, we also compare our scheme with the Costas loop, because it has also been a well-researched and popular technique for carrier recovery. The block diagram for a Costas loop is shown in Fig. 4.

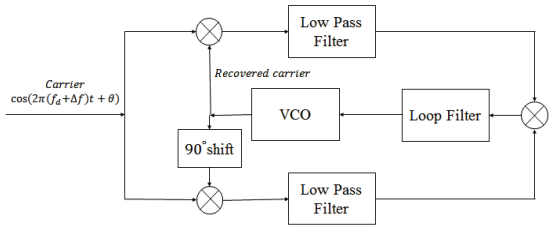


Fig. 4. Block diagram of a Costas loop

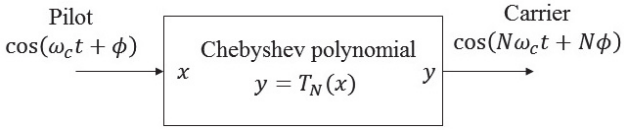


Fig. 5. Block diagram of the proposed method

III. Description of Scheme

The Chebyshev polynomials follow the general relationship

$$T_n(x) = 2xT_{n-1}(x) - T_{n-1}(x) \tag{5}$$

where $T_0(x) = 1$ and $T_1(x) = x$. These polynomials can be used to represent any harmonic of a cosine as a polynomial of the cosine. For example, $n = 2$ and $x = \cos(\theta)$ give the well known trigonometric identity

$$cos(2\theta) = 2cos^2(\theta) - 1 \tag{6}$$

Given any carrier that is a harmonic of a recoverable pilot tone, the Chebyshev polynomial can be used to express the carrier as a polynomial in the pilot, which enables the recovery of the carrier from the pilot without the need for a control loop and without the need for parameter tuning. The most expensive operation of this method would be the generation of the highest order term of the polynomial. For example, if the carrier is the third harmonic, a cuber block would be required. However, this cost would need to be borne by conventional schemes as well. Hence non-idealities in the block that generates the higher order terms will be unavoidable and common for all schemes. Hence the comparison to be made would be in speed and accuracy.

This method is guaranteed to generate the exact frequencies and phases of the carriers, and also performs the recovery instantaneously. It can be seen that parameter tuning as in the case of a control loop is not required, and there is no upper limit on the frequency or phase deviation that the scheme can track in finite time, unlike in a control loop. The blocks in the diagram are easily realisable with modern digital hardware, and even more so with Software Defined Radio (SDR), which offers precision and flexibility. Protocols crafted with this particular method in mind can be used for fast and accurate burst transmissions, and the practical implications of the method are significant.

The reason this method is advantageous is that it takes advantage of the special relationship between the pilot and

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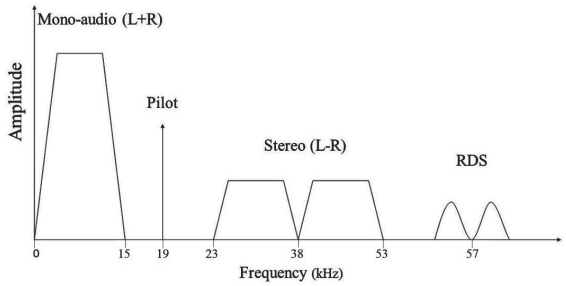


Fig. 6. Schematic of amplitude spectrum of FM-RDS broadcast

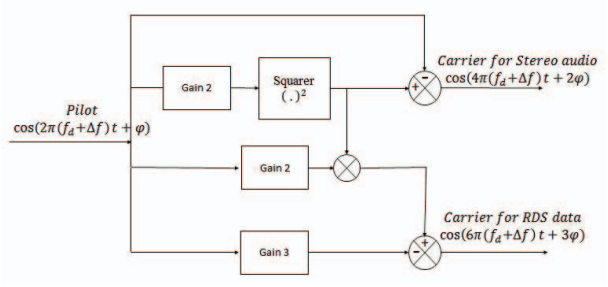


Fig. 7. Block diagram of proposed carrier recovery scheme for FM-RDS

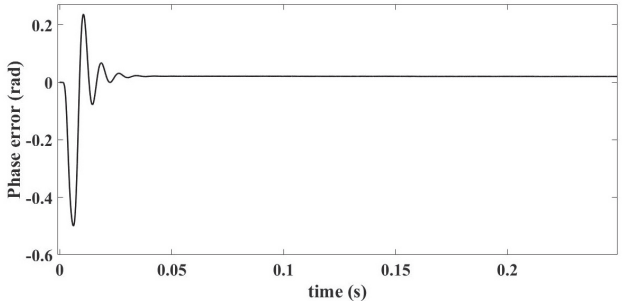


Fig. 8. Error in the PLL carrier recovery scheme as a function of time

the carriers in protocols that have such a relationship, while the PLL and Costas loop are more generic methods that are applied for such protocols without regard for the harmonic relationship. The scheme is faster at recovering the carriers because it does not rely on control-loop feedback and stabilization, since the trigonometric relations are identities that guarantee the faithful and immediate reproduction of frequency and phase offsets in the pilot in the recovered carriers. Frequency-domain filters are unnecessary after the extraction of the pilot tone. Only simple multiplication, addition, and subtraction operations are made use of, and thus the scheme is cheaper to implement as well. The blocks used to generate the higher order terms are the same as in the control loop schemes, and hence the terms of the Chebyshev polynomial are generated already by control loop schemes, but instead of using a bandpass filter and then the control loop, the carrier is recovered with simple arithmetic applied to the polynomial terms of the Chebyshev expression applied to the pilot.

IV. Experiments and Results

A second order PLL, simulated in software, with loop parameters $\alpha = 0.05$ and $\beta = 0.01$, is used for comparison with the proposed scheme. A sinusoidal phase detector is used, with a VCO with voltage to frequency transfer coefficient of 3.8 kHz/V for stereo carrier recovery. The parameters satisfy the stability criteria for a second order discrete PLL [8]. The specific case of the FM-RDS protocol is used, which has a pilot tone at 19 kHz and carriers at 38 kHz and 54 kHz, which are frequency locked second and third harmonics of the pilot. A schematic of the amplitude spectrum of the FM-RDS scheme is shown in Fig. 6.

The corresponding Chebyshev polynomials used to generate the carriers for the stereo audio and the RDS data would be of orders $n = 2$ and $n = 3$ respectively. The 38 kHz carrier is generated as

$$cos(2\theta) = 2cos^2(\theta) - 1 \tag{7}$$

and the 57 kHz carrier is generated as

$$cos(3\theta) = 4cos^3(\theta) - 3cos(\theta) \tag{8}$$

where $cos(\theta)$ is the pilot signal at 19 kHz. These carriers can thus be generated by the block diagram shown in Fig. 7.

Pilot signals are simulated on MATLAB® as cosines at a frequency of 19 kHz, with frequency offsets and phase offsets. Fig. 8 shows the error signal of the PLL as a function of time, for a pilot signal with frequency deviation of 2 Hz, which is the tolerance limit for frequency deviation for the pilot according to the RDS standard [9]. The three schemes obtain the stereo audio carrier at 38.004 kHz, but the proposed scheme does so instantaneously, without requiring an error stabilisation time. Within this time, any data that is received will be attenuated or distorted. The proposed scheme is much less complex and takes lesser execution time, and can hence recover data accurately from the start of transmission itself. This is ideal for the transmission and recovery of bursts of information.

Similarly, a 57 kHz RDS digital data carrier with a phase deviation of $\frac{3\pi}{20}$ was simulated and fed to a Costas loop, having a voltage to frequency conversion factor (δ) of 1.2Hz/V. The predicted phase offset of the carrier, over time, for a Costas loop carrier recovery system is shown in Fig. 9. There is finite delay before the phase stabilises at $\frac{3\pi}{20}$.

Fig. 10 shows the time domain signals of RDS 57 kHz carriers recovered by the proposed method, and the ideal recovered carrier, for a pilot phase deviation of $\frac{\pi}{20}$. Since the carrier is the locked third harmonic of the pilot, the ideal carrier is simulated as a 57 kHz cosine with a phase offset of $\frac{3\pi}{20}$. It is seen that the ideal recovered carrier coincides with the one generated by the proposed scheme.

A more objective analysis is presented in Table I, showing settling times, and in Table II, showing mean error in phase over a time period of 6.67 seconds for the three schemes, for a phase deviation of $\frac{\pi}{20}$ in the pilot, leading to a phase deviation

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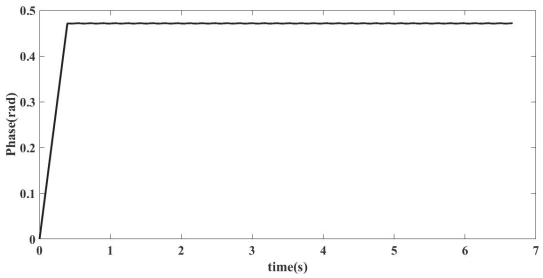


Fig. 9. Predicted phase of recovered carrier vs. time using a Costas loop

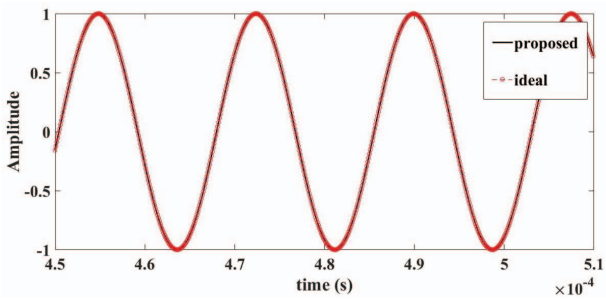


Fig. 10. Recovered 57 kHz carrier in the time domain for a pilot phase deviation of $\frac{3\pi}{20}$ and the ideal 57 kHz carrier.

of $\frac{3\pi}{20}$ in the 57 kHz RDS carrier. Additive white Gaussian noise was added to the pilot signal with a signal-to-noise ratio (SNR) of 20 dB to represent noise in the channel. The reported values are heavily dependent on the choice of parameters used in the phase-locked loops and the Costas loops filters and step-update factors, and hence these parameters are also mentioned for reproducibility of results. It must be noted that the squarer and cuber blocks are required for both control loop schemes and the proposed method, and hence contribute to the settling time equally for both. The difference arises after these blocks, which means that any non-idealities associated with the generation of the Chebyshev polynomials due to the higher order terms must and will be associated with the control loop schemes as well.

Settling time is defined as the time required for the response curve to reach and stay within a range of 2% of the final value [10]. A sampling frequency of 150 kHz was used for the software simulation. It is to be noted that different values of the parameters in the PLL and in the Costas loop will give different settling times and accuracies. This is inherently disadvantageous compared to the proposed scheme, that has no parameters to be tuned by the designer. We have presented results of experiments run with two sets of parameters for each competing recovery scheme.

It can be seen that the proposed method does better than both the PLL and the Costas loop in terms of settling time as well as mean phase error. The settling time is an indicator of how long it will take for data to start being received correctly, and the mean phase error is an estimate of the corruption in the signal over the short time before it stabilizes. Longer

TABLE I
SETTLING TIMES

Method	Parameters	Settling time (s)
PLL	$\alpha = 0.05, \beta = 0.01$ rad	0.0145
PLL	$\alpha = 0.08, \beta = 0.02$ rad	0.033
Costas	$\delta = 1.2$ Hz/V	0.385
Costas	$\delta = 0.5$ Hz/V	0.924
Proposed	-	0

TABLE II
MEAN PHASE ERRORS

Method	Parameters	Mean Phase error (rad)
PLL	$\alpha = 0.05, \beta = 0.01$ rad	0.0002
PLL	$\alpha = 0.08, \beta = 0.02$ rad	0.0042
Costas	$\delta = 1.2$ Hz/V	-0.1430
Costas	$\delta = 0.5$ Hz/V	-0.1642
Proposed	-	0.0001

settling times naturally cause a delay in data reception, and the proposed method is able to surmount this difficulty and enable instantaneous data reception even in the presence of noise in the channel.

Squaring and cubing are operations that are required in all three recovery schemes and hence are a part of the baseline in terms of complexity as well as the effect of noise. The proposed scheme is able to do better in spite of the squaring or cubing of the pilot with AWGN, which further spreads the spectrum of the noise and introduces other spurious frequencies, and has a lower mean error in phase than the other schemes. Lower error in the phase of the recovered carrier directly translates to cleaner data recovery. This indicates the robustness of the proposed scheme against noise while still being instantaneous in recovery.

V. DISCUSSION

The proposed scheme can recover the exact carriers for any frequency and phase offset in the carrier used on the transmission side, as long as the carrier in the protocol is a harmonic of a recoverable pilot. This scheme can be used effectively even when the information is to be transmitted in very short bursts, as the recovery is instantaneous, as well as in the presence of noise. In a conventional scheme, until steady state is reached information received is distorted. PLL carrier recovery and Costas loop carrier recovery are general schemes that can work for various protocols, but the proposed scheme is better suited for protocols as described earlier, as it takes advantage of the special relationship between the pilot and the carriers. The method requires only digital multipliers, adders, and subtractors, and is hence cost-efficient as well.

VI. CONCLUSION

The superiority of the proposed scheme to conventional PLL and Costas loop carrier recovery schemes is shown. The scheme can be used in any protocol that has carriers that are harmonics of a pilot. Higher order harmonics can be generated as well using the Chebyshev recursive formulation, though we have shown results for the specific case of the second and third

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harmonics only. The results clearly show the potential of the scheme for accurate carrier and data recovery. The scheme is easily implementable on modern digital hardware and in software defined radio. In a conventional implementation, the existing analog methods namely power law devices followed by phase locked loops or Costas loops are implemented in the discrete domain which still have legacy issues like transient response and stability. Using simple trigonometric identities, we have shown that it is possible to have novel real time and accurate implementations of tested methods which also can alleviate the legacy problems associated with the original analog designs.

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